$$\frac{f(x+\Delta x)-f(x)}{\Delta x} = a^x\left(\frac{a^{\Delta x}-1}{\Delta x}\right).$$

We know that $\lim_{\Delta x\to 0}\frac{a^{\Delta x}-1}{\Delta x} = \log a$.

Therefore, $\lim_{\Delta x\to 0}\frac{f(x+\Delta x)-f(x)}{\Delta x} = a^x \lim_{\Delta x\to 0}\left(\frac{a^{\Delta x}-1}{\Delta x}\right)$

$$= a^x \times \log a$$

or $\boxed{\frac{d}{dx}(a^x) = a^x \log a}$.

In particular, $\frac{d}{dx}(e^x) = e^x \log e$

$$\boxed{\frac{d}{dx}(e^x) = e^x}.$$

(5) The derivatives of the Trigonometric functions

(i) The sine function, sin*x*.

Let $y = f(x) = \sin x$.

Then $f(x+\Delta x) = \sin(x+\Delta x)$ and

$$f(x+\Delta x)-f(x) = \sin(x+\Delta x)-\sin x = 2\sin\frac{\Delta x}{2}\cos\left(x+\frac{x}{2}\right).$$

Now $$\frac{f(x+\Delta x)-f(x)}{\Delta x} = \frac{\sin\left(\frac{\Delta x}{2}\right)}{\left(\frac{\Delta x}{2}\right)}\cos\left(x+\frac{\Delta x}{2}\right).$$

This implies, $$\lim_{\Delta x\to 0}\frac{f(x+\Delta x)-f(x)}{\Delta x} = \lim_{\Delta x\to 0}\frac{\sin\left(\frac{\Delta x}{2}\right)}{\left(\frac{\Delta x}{2}\right)}.\lim_{\Delta x\to 0}\cos\left(x+\frac{\Delta x}{2}\right)$$

$= 1 \times \cos x$ (Since $\cos x$ is continuous, $\lim_{\Delta x\to 0}\cos(x+\Delta x) = \cos x$) .

$= \cos x$

That is $\boxed{\frac{d}{dx}(\sin x) = \cos x}$.

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